

Question number	Answer	Additional guidance	
2(a)(i)	Choose an item. • B is the correct answer A is not the correct answer as cytoplasm is not the site of the ETC C is not the correct answer as mitochondrial matrix is not the site of the ETC D is not the correct answer as outer mitochondrial membrane is not the site of the ETC		Computer (1)

Question number	Answer	Additional guidance	
2(a)(ii)	Choose an item. D is the correct answer A is not the correct answer as a carbohydrate is not an enzyme in the production of ATP B is not the correct answer as a lipid is not an enzyme in the production of ATP C is not the correct answer as a phospholipid is not an enzyme in the production of ATP		Computer (1)

Question number	Answer	Additional guidance	
2(a)(iii)	A description that includes the following points: { transports / carries / provides } { hydrogen (atoms) / electrons / hydrogen ions } (to ETC / inside mitochondria) (1) (hydrogen ions / electrons) result in { release of energy / ATP production } (in ETC) (1)	accept protons for hydrogen ions accept for chemiosmosis to occur / to produce protons to be pumped to intermembrane space / for oxidative phosphorylation to occur / description of chemiosmosis ignore production of energy	Expert (2)

Question number	Answer	Additional guidance	Mark
2(b)(i)	An answer that includes the following points: • less ATP produced (1) • (because) haemoglobin cannot transport (as much) oxygen / {prevents / reduces} oxygen binding to haemoglobin (1) • (therefore) so cells have {no / less} final electron acceptor (1)	may give numbers 2:36 accept binds to haemoglobin instead of oxygen accept haemoglobin has a higher affinity for CO than oxygen accept aerobic respiration can't occur / less aerobic respiration do not accept anaerobic respiration on its own – it needs linking to ATP accept (more) anaerobic respiration occurs which produces fewer ATP molecules (per glucose) for 2 marks	Expert (3)

Question number	Answer	Additional guidance	Mark
2(b)(ii)	An explanation that includes the following points: • anaerobic respiration involves (just) glycolysis / ATP produced during glycolysis (1) • anaerobic respiration occurs in cytoplasm / anaerobic respiration doesn't involve the {mitochondria/ETC}	accept substrate level phosphorylation for glycolysis accept ATP is produced when glucose is converted to pyruvate	Expert (2)

